



DATABASES

AN INTRODUCTION



OVERVIEW

- What is a database?
- What is a relational database?
 - Types of relationships
 - Keys
- What is an ERD? How is it made?
- Activity: Make an ERD
- What is SQL and Postgresql?
 - Creating Tables
 - Data types

WHAT IS A DATABASE?

- We have all heard of databases and we know they store some kind of data...
- By definition, a database is an organized collection of structured information (or data), typically stored electronically in a computer system.
- Database structure is based on the idea of organizing data in Tables
- We use an RDBMS to create and interact with our database tables.

First Name	Last Name	Address	City	Age
Mickey	Mouse	123 Fantasy Way	Anaheim	73
Bat	Man	321 Cavern Ave	Gotham	54
Wonder	Woman	987 Truth Way	Paradise	39
Donald	Duck	555 Quack Street	Mallard	65
Bugs	Bunny	567 Carrot Street	Rascal	58
Wiley	Coyote	999 Acme Way	Canyon	61
Cat	Woman	234 Purrfect Street	Hairball	32
Tweety	Bird	543	Ittittaw	28

WHAT IS AN RDBMS?

- Relational DataBase Management System
- This provides a place to store our tables, divided into schemas, and stored in databases
- The RDBMS provides compilation and optimization of queries behind the scenes.
- We will be using PostgreSQL, a free and open-source RDBMS

WHAT IS A RELATIONAL DATABASE?

- By definition, a relational database is a database structured to recognize relations among stored items of information.
- These relationships are how we link multiple tables together
- We have three types of relationships:
 - One-to-one
 - One-to-many or Many-to-one
 - Many-to-many

TYPES OF RELATIONSHIPS

- One-to-one relationship
 - The least used type of relationship
 - Used when both tables only relate to one row in the other
 - For example, one student will only have one identification card and any one identification card can only belong to one student.
- One-to-Many relationship
 - The most used type of relationship
 - Used when one table can relate to many rows in the other, but not the other way around.
 - For example, one school can have many students and any one student should belong to only one school.

TYPES OF RELATIONSHIPS

- Many-to-Many
 - This is used when both tables could have multiple rows on the other side.
 - For example, one student can have many classes and any one particular class can have many students.
- The relationship between two tables is known as Multiplicity

HOW DO WE MAKE RELATIONSHIPS?

- Relationships are defined using keys
- There are many types of keys:
 - Primary Key: the main identifier of a row. Must be unique
 - Candidate Key: a key that uniquely identifies the row, but may not be the primary key
 - Foreign Key: a key that is the primary key in another relation(table)
 - Natural Key: a key which comes from the data, like email or license plate
 - Surrogate Key: a key which does not come from the data, but is generated for the sole purpose of being an id

WHAT IS AN ERD?

- ERD stands for Entity Relationship Diagram
- It is a way to visualize data tables and the relationships between them
- There are many free tools for creating an ERD such as draw.io and GitMind

ERD EXAMPLE

CHALLENGE

- Try making an ERD for a store that contains an employee table, customer table, and orders table.
- Each table must have a primary key.
- Each order must relate to the customer who placed the order, and the employee who took the order.

WHAT IS SQL?

- SQL stands for Structured Query Language which is the language used by databases
- PostgreSQL is one of the many dialects of SQL
- To create a database, we will create an ElephantSQL account and start a tiny turtle(free) hosted database

CREATING TABLES

- We can create tables by running queries
- For example:

```
CREATE TABLE Students (  
  StudentId INTEGER PRIMARY KEY,  
  FirstName VARCHAR(100),  
  LastName VARCHAR(100),  
);
```

INSERTING DATA

- We can insert real data into our tables using the INSERT statement
- This statement adds 2 rows to the student table
 - `INSERT INTO Student(StudentId, FirstName, LastName) VALUES (1, Tom, Riddle), (2, Ron, Weasley);`

DATA TYPES

- Char(n) are fixed-length strings where n is the length
- Varchar(n) are fixed-length and blank padded strings
- Integer and other Java analogues exist, ex: ShortInt, BigInt
- Numeric to store whole and decimal numbers with a given decimal precision
- Date stores date information
- Timestamp stores time and date information

QUERY KEYWORDS

- SELECT starts the data retrieval query
- FROM tells the DB which table to target
- WHERE is a boolean condition that all rows in the result set must conform to
- GROUP BY works only with functions like SUM, and allows for subgroupings of results based on given columns
- ORDER BY tells the DB what order to display the given columns in. ASC is ascending order while DESC is descending order

RETRIEVING DATA

- This query selects all rows from the Student table
 - `SELECT * FROM Student ;`
- This query selects the first and last name data fields where the student id is 2
 - `SELECT FirstName, LastName FROM Student WHERE StudentId = 2;`

CHALLENGE 2

1. Create the following tables and their keys with the given fields:
 - Residents: id, first_name, last_name, apartment_id
 - Cars: id, make, model, year, license_plate, owner_id
 - Apartments: id, building_letter, room_number, monthly_rent
 - Pets: id, breed, name, apartment_id, is_service_animal
2. Insert 3 rows of dummy data into each table
3. Construct the following SELECT queries
 - Get all residents
 - Get all residents in building B
 - Get the average rent in building C
 - Get the total number of pets